Multi-agentic implementation of a 2D topology optimization framework for compliance and linearized buckling via weighted sum method

Gabriele Ricatto

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Abstract

This paper presents a framework for multi-objective topology optimization, focusing on the trade-off between structural compliance and linear buckling stability. A custom Finite Element Method solver, based on $\mathcal{Q}_4$ bilinear isoparametric elements, is implemented. The material distribution is parameterized utilizing the Solid Isotropic Material with Penalization (SIMP) approach. To explore the trade-off spectrum between global stiffness and the fundamental Buckling Load Factor (BLF), a weighted and dynamically normalized sensitivity scheme is implemented. The proposed framework provides an effective tool for generating preliminary components, illustrating transitions across the design space when stiffness is penalized in favor of structural stability and vice versa. Furthermore, a comparison between the results obtained and those derived from commercial software is presented.

The low-level debugging and code writing were primarily conducted using a multi-agentic human-in-the-loop workflow, leveraging commercially available large language models through Google's Antigravity.

1 Introduction

The goal of this work is the topology optimization of a two-dimensional plane-stress domain subjected to in-plane mechanical loads, considering both structural compliance and linearized buckling resistance [5, 6]. The benchmark problem is a rectangular cantilever beam of length L and height H, clamped along its left vertical edge ($u_x = u_y = 0$) and loaded by a downward concentrated vertical force F at the bottom-right corner. The material distribution is sought within a fixed reference mesh under a prescribed volume fraction constraint V_f to study how the layout changes when stability is weighted alongside stiffness.

2 FEM implementation

For the implementation of the FEM solver, we rely on bilinear isoparametric elements ($\mathcal{Q}_4$) applied to a fully structured mesh. The rectangular elements are aligned with the global reference system, with the origin of the natural coordinates centered in the centroid of each element. The general formulation for variable-size elements has been omitted, as unnecessarily complicated for the purpose of this project. In accordance with the simplifications introduced, the mapping between physical and natural coordinates assumes the following form:

$$x = x_1 + \frac{dx}{2}\xi, \quad y = y_1 + \frac{dy}{2}\eta \tag{1}$$

with $\eta \in [-1; 1]$ and $\xi \in [-1; 1]$. We can now derive the Jacobian as:

$$[J] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} = \begin{bmatrix} dx/2 & 0 \\ 0 & dy/2 \end{bmatrix} \implies \det(J) = \frac{1}{4} dx dy \tag{2}$$

and its inverse:

$$[J]^{-1} = \begin{bmatrix} \frac{2}{dx} & 0 \\ 0 & \frac{2}{dy} \end{bmatrix} \tag{3}$$

The shape functions are:

$$N_1 = \frac{1}{4}(1-\xi)(1-\eta) \tag{4}$$

$$N_2 = \frac{1}{4}(1+\xi)(1-\eta) \tag{5}$$

$$N_3 = \frac{1}{4}(1+\xi)(1+\eta) \tag{6}$$

$$N_4 = \frac{1}{4}(1-\xi)(1+\eta) \tag{7}$$

The writing and calculation of the elementary stiffness matrix K_e^0 are quite simple and are done only once at the beginning of the calculation. After that,

in accordance with the SIMP model, at each iteration the following calculation is foreseen:

$$[K] = \sum_{e=1}^{m} E_K(\hat{x}_e) \cdot K_e^0 = \sum_{e=1}^{m} [E_{min} + (E_0 - E_{min})\hat{x}_e^{p_K}][K_e^0] \tag{8}$$

We have already introduced the parameters p_K and p_G, but we have not yet assigned a value to it. In accordance with numerical evidence and those found in literature, we opt to assign both a value equal to 3, so $p_K = p_G = p = 3$ [1]. As for the elementary matrix K_e^0:

$$[K_e^0] = \int_V [B]^T [C][B] dV \tag{9}$$

The element is two-dimensional; thus, assuming a unit thickness, the elemental stiffness matrix is defined as:

$$[K_e^0] = \int_\Omega [B]^T [C][B]\, dxdy \tag{10}$$

where the matrix $[B]$ contains the spatial derivatives of the shape functions. By transitioning to natural coordinates to resolve the integral, the formulation becomes:

$$[K_e^0] = \int_{-1}^{1}\int_{-1}^{1} [B(\xi,\eta)]^T [C][B(\xi,\eta)] \det(J)\, d\eta d\xi \tag{11}$$

The elemental geometric stiffness matrix, G_e^0, is derived from the linearization of the Principle of Virtual Work:

$$[G_e^0] = \int_{\Omega_0} B_1^T T[\sigma_e] B_1 d\Omega_0 \tag{12}$$

where for a Q4 element the matrix B_1 is composed of the gradients of the shape functions, and furthermore we have the stress matrix:

$$T[\sigma_e] = \begin{bmatrix} \sigma_x & \tau_{xy} & 0 & 0 \\ \tau_{xy} & \sigma_y & 0 & 0 \\ 0 & 0 & \sigma_x & \tau_{xy} \\ 0 & 0 & \tau_{xy} & \sigma_y \end{bmatrix} \tag{13}$$

Where for simplicity, before optimization, the constant elements are pre-calculated by collecting the individual stresses in binary matrices:

$$T[\sigma_e] = \sigma_x \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \sigma_y \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \tau_{xy} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{14}$$

The objective of the decomposition proposed is to separate the state-dependent variables (local stresses) from the geometric constant terms. By substituting the

decomposed stress matrix back into the integral, the scalar stress components can be extracted from the integral as follows:

$$[G_e^0] = \sigma_x \int_{\Omega_0} B_1^T T_x B_1 \, d\Omega_0 + \sigma_y \int_{\Omega_0} B_1^T T_y B_1 \, d\Omega_0 + \tau_{xy} \int_{\Omega_0} B_1^T T_{xy} B_1 \, d\Omega_0 \quad (15)$$

where T_x, T_y, and T_{xy} denote the constant basis matrices presented in (14). This mathematical arrangement yields a significant computational advantage since the three integral terms now depend exclusively on the element geometry and the shape functions, which remain invariant throughout the optimization process on a fixed mesh. Consequently, these matrices can be evaluated exactly once during the pre-processing phase: by doing so, within the optimization loop they will simply be scaled by the updated stress values at each iteration.

To give a better insight of the FEM workflow, in the following one can find the steps followed by the solver:

1. For each element in the mesh, K_e^0 is evaluated. This computation is performed only once during the pre-processing phase. With the results obtained, the global stiffness matrix $[K]$ is assembled by penalizing the elemental matrices according to the SIMP interpolation scheme, $K_e = E_K(\hat{x}_e)K_e^0$.

2. Boundary conditions (loads and constraints) are applied and the system $KU = F$ is solved.

3. Having retrieved the global displacement vector U, the overall structural compliance is evaluated as $C = U^T KU$.

4. The assembly of the geometric stiffness matrix G requires the local stress state of each element. Extracting the local nodal displacements u_e from the global vector U, the elemental strains and stresses are computed using $\epsilon_e = Bu_e$ and $\sigma_e = D\epsilon_e$.

5. The linear buckling eigenvalue problem $[K + \lambda_i G]\{\phi_i\} = \{0\}$ is solved to extract the fundamental buckling load factors λ and their corresponding eigenmodes ϕ. These results will then be used by the optimizer.

3 Optimization problem setup

In accordance with [7], the spatial domain is discretized in m equal elements for a total of n *Degrees of Freedom* (DOFs). We also define the pseudo-density field as $\hat{x} = \{\hat{x}_e\}_{e=1}^m$, that are partitioned between 3 sets:

1. $\Omega_{p_0} \rightarrow$ passive regions where $\hat{x}_e = 0$ (forced void)

2. $\Omega_{p_1} \rightarrow$ passive regions where $\hat{x}_e = 1$ (forced solid)

3. $\Omega_A \rightarrow$ actual domain with active design space variables.

In the design space, density $\hat{x}_e$ is obtained through a smoothed Heaviside projection [8]:

$$\hat{x}_e = \frac{\tanh(\beta\eta_H) + \tanh(\beta(\tilde{x}_e - \eta_H))}{\tanh(\beta\eta_H) + \tanh(\beta(1 - \eta_H))} \tag{16}$$

with threshold $\eta_H \in [0, 1]$ (fixed at $\eta_H = 0.5$) and sharpness $\beta \in [1, \infty)$. This equation, using the hyperbolic tangent, forces the intermediate values towards 0 and 1, in order to decrease the shades and define sharper boundaries.
The intermediate variable $\tilde{x}_e$ represents the filtered density, calculated via:

$$\tilde{x}_e = \frac{\sum_{i=1}^{m} h_{ei} x_i}{\sum_{i=1}^{m} h_{ei}} \tag{17}$$

with $h_{ei} = \max(0, r_{min} - dist(\Omega_i, \Omega_e))$. In this expression, r_{min} specifies the filter radius, and $\text{dist}(\Omega_i, \Omega_e)$ represents the center-to-center distance between elements i and e. The introduction of the previous formulation is needed in order to exclude elements further than r_{min} from an element e, thus increasing the weight of elements close to e. Ultimately, these measures prevent the generation of mesh-dependent solutions and checkerboarding patterns.

Let the Young's modulus be parametrized by the SIMP interpolations with penalizations $p_K, p_G > 1$:

$$E_K(\hat{x}_e) = E_{min} + (E_0 - E_{min})\hat{x}_e^{p_K} \tag{18}$$

$$E_G(\hat{x}_e) = E_0 \hat{x}_e^{p_G} \quad \text{(see (39))} \tag{19}$$

where E_0 represents the elastic modulus of the solid base material, and $E_{min} \ll E_0$ is a positive lower bound (in this case, set to $10^{-9}E_0$).

The volume fraction is defined as $f(\hat{x}) = \frac{1}{m}\sum_{e=1}^{m} \hat{x}_e$. Alongside this we introduce the compliance $C(\hat{x}) = F^T U(\hat{x})$, which represents the external work done by the loads: minimizing this compliance is equivalent to maximizing the global stiffness of the design.

To predict structural instability while maintaining computational efficiency, a linearized buckling analysis is adopted. This requires solving the following generalized eigenvalue problem:

$$(K + \lambda_i G)\phi_i = 0 \tag{20}$$

Specifically, the fundamental Buckling Load Factor (BLF), denoted as λ_1, is retrieved through the minimization of:

$$\lambda_1(\hat{x}, U) = \min_{v \in \mathbb{R}^n, v \neq 0} \frac{v^T K(\hat{x}) v}{-v^T G(U) v} \tag{21}$$

where v represents a non-zero virtual displacement field. Only positive BLFs, which indicate structural instability under the specific applied load direction, are considered. In computational practice, to prevent numerical singularities,

the solver calculates the reciprocal eigenvalue $\mu_i = 1/\lambda_i$. Consequently, the problem can be formulated as:

$$(G(U) + \mu_i K(\hat{x}))\phi_i = 0 \tag{22}$$

As demonstrated by Ferrari and Sigmund [1, 7], the lowest buckling eigenvalues often coalesce and switch order during optimization. To avoid non-differentiable mode switching, the first n_{Eig} reciprocal buckling eigenvalues are aggregated using the Kreisselmeier-Steinhauser (KS) function [7] normalized by the fundamental value μ_1:

$$\mu_{KS} = \mu_1 \left[1 + \frac{1}{\rho_{KS}} \ln \sum_{m=1}^{n_{Eig}} \exp\left(\rho_{KS} \frac{\mu_m - \mu_1}{\mu_1} \right) \right] \tag{23}$$

where $\rho_{KS} > 0$ is the aggregation parameter. To explore the conflicting criteria between structural compliance and linear buckling resistance, the two objectives can be combined via a weighted sum method [3]:

$$\tilde{f}(\hat{x}) = \sum_{i=1}^{M} w_i f_i(\hat{x}) \tag{24}$$

where $w_i \in [0, 1]$ represents the individual weighting parameter assigned to the i-th objective. In nominal form, the scalar objective governed by the parameter $\alpha \in [0, 1]$ reads:

$$F(\vec{x}) = \alpha \mu_{KS} + (1 - \alpha) C \tag{25}$$

Combining those two directly into a single scalar function is problematic, as their respective raw sensitivities differ by several orders of magnitude. A direct weighted sum would cause the objective with the steeper local gradients to dominate the density update.

In practice, the solver does not minimize (25) directly, but operates on normalized sensitivities at each iteration (i) through (27). Rather than using the arithmetic mean (which is sensitive to localized stress peaks near the point load), both gradient fields are scaled by their 95-th percentile absolute value (P_{95}):

$$S_C^{(i)} = P_{95}\left(\left| \frac{\partial C^{(i)}}{\partial \hat{x}} \right| \right), \quad S_\mu^{(i)} = P_{95}\left(\left| \frac{\partial \mu_{KS}^{(i)}}{\partial \hat{x}} \right| \right) \tag{26}$$

Dividing the raw gradients by these factors brings the two sensitivities to the same order of magnitude at each step. The search direction passed to the optimizer is the linear combination governed by the effective weight $\alpha_{eff}^{(i)}$:

$$\frac{\partial F^{(i)}}{\partial \hat{x}_e} = (1 - \alpha_{eff}^{(i)}) \frac{1}{S_C^{(i)}} \frac{\partial C^{(i)}}{\partial \hat{x}_e} + \alpha_{eff}^{(i)} \frac{1}{S_\mu^{(i)}} \bar{g}_{\mu,e}^{(i)} \tag{27}$$

where $\bar{g}_{\mu,e}^{(i)}$ is the clamped buckling sensitivity defined in Section 7.

Classical scalarization theory [11] states that minimizing a convex combination with fixed, positive weights yields Pareto-optimal points, although a linear weighted sum cannot reach non-convex regions of the front [9, 10]. In our algorithm, however, the scaling factors $S_C^{(i)}$ and $S_\mu^{(i)}$ are recomputed at every iteration, $\alpha_{eff}^{(i)}$ is ramped, positive buckling sensitivities are truncated ($\bar{g}_{\mu,e}$), and β is updated by continuation. For these reasons, the sweep over α is a practical trade-off study across design alternatives rather than an exact Pareto front, and dominated points can appear when modes switch.

Furthermore, α is not a raw weight on μ_{KS}: to keep a continuous load path and avoid disconnecting the tension chord, α is capped at an effective value $\alpha_{eff} \leq 0.65$ (see Section 7). Setting $\alpha = 1$ therefore produces a compromise that retains a 35% compliance contribution. In the dashboard, α is varied from 0 to 1 in steps of 0.1 to track how the layout changes as buckling weight increases.

4 Sensitivity Analysis

To evaluate the minimum of λ, and therefore the max of μ, we calculate the following derivative [4]:

$$\frac{\partial \mu_i}{\partial \hat{x}_e} = -\left[\phi_i^T \frac{\partial G}{\partial \hat{x}_e}\phi_i + \mu_i \phi_i^T \frac{\partial K}{\partial \hat{x}_e}\phi_i - w_i^T \frac{\partial K}{\partial \hat{x}_e} U\right] \tag{28}$$

where:

- ϕ_i: eigenvector / buckling mode shape.
- μ_i: inverse of the critical load factor.
- U: displacement vector.
- w_i: adjoint vector. It is a vector of virtual displacements, obtained by solving $w_i = K^{-1}[\phi_i^T(\nabla_U G)\phi_i]$, where ∇_U represents the gradient with respect to the displacements U.

As for the elemental matrices derivatives, we can write:

$$\frac{\partial K^e}{\partial \hat{x}_e} = \frac{\partial E_K(\hat{x}_e)}{\partial \hat{x}_e} K_0^e, \quad \frac{\partial G^e}{\partial \hat{x}_e} = \frac{\partial E_G(\hat{x}_e)}{\partial \hat{x}_e} G_0^e \tag{29}$$

while the compliance derivative is evaluated as [7]:

$$\frac{\partial C}{\partial \hat{x}_e} = -U_e^T K_0^e U_e \frac{\partial E_K(\hat{x}_e)}{\partial \hat{x}_e} \tag{30}$$

5 Updating process

For the material update at each iteration i, an Optimality Criteria (OC) scheme is employed [2]. When the Heaviside projection and density filter are active, the

sensitivities with respect to the design variables x_e are obtained through the chain rule described in Section 6.

For the update rule:

$$x_e^{new} = \begin{cases} \max(x_{min}, x_e - \zeta) & \text{if } x_e B_e^{\eta_{OC}} \leq \max(x_{min}, x_e - \zeta) \\ x_e B_e^{\eta_{OC}} & \text{if } \max(x_{min}, x_e - \zeta) < x_e B_e^{\eta_{OC}} < \min(1, x_e + \zeta) \\ \min(1, x_e + \zeta) & \text{if } \min(1, x_e + \zeta) \leq x_e B_e^{\eta_{OC}} \end{cases} \tag{31}$$

with initial move limit $\zeta = 0.2$ and initial damping exponent $\eta_{OC} = 1/2$ (both reduced during β-continuation, see Section 6). x_{min} is fixed at 0.001 to avoid numerical singularities. Finally, B_e is defined as follows:

$$B_e = \frac{-\frac{\partial \tilde{F}}{\partial x_e}}{\lambda_L \frac{\partial \tilde{V}}{\partial x_e}} \tag{32}$$

where $\frac{\partial \tilde{F}}{\partial x_e}$ and $\frac{\partial \tilde{V}}{\partial x_e}$ denote the filtered sensitivities of the objective and projected volume fraction, and λ_L is the Lagrangian multiplier.

The Lagrangian multiplier λ_L is determined through the bisection method, so that the limit on the volume is satisfied:

$$\lambda_L = 0.5(l_2 + l_1); \quad l_2^{(0)} = 10^9, \quad l_1^{(0)} = 0 \tag{33}$$

The algorithm stops when $(l_2 - l_1)/(l_2 + l_1) < 10^{-4}$, and the update follows the classic bisection procedure.

6 Projection chain rule and continuation scheme

With the Heaviside projection (16) and the density filter (17), the sensitivities computed with respect to the physical densities $\hat{x}_e$ are mapped back to the design variables x_i by the chain rule. Differentiating (16) with respect to the filtered density $\tilde{x}_e$ gives:

$$\frac{\partial \hat{x}_e}{\partial \tilde{x}_e} = \frac{\beta \left(1 - \tanh^2(\beta(\tilde{x}_e - \eta_H))\right)}{\tanh(\beta \eta_H) + \tanh(\beta(1 - \eta_H))} \tag{34}$$

Applying the discrete transpose of the spatial filter (17), the sensitivities of the objective and volume fraction with respect to x_i are:

$$\frac{\partial \tilde{F}}{\partial x_i} = \sum_{e=1}^{m} \frac{h_{ei}}{\sum_{k=1}^{m} h_{ek}} \frac{\partial \hat{x}_e}{\partial \tilde{x}_e} \frac{\partial F}{\partial \hat{x}_e}, \quad \frac{\partial \tilde{V}}{\partial x_i} = \sum_{e=1}^{m} \frac{h_{ei}}{\sum_{k=1}^{m} h_{ek}} \frac{\partial \hat{x}_e}{\partial \tilde{x}_e} \tag{35}$$

A continuation scheme is used for the projection parameter β. Starting from $\beta = 1$, β is doubled every 20 iterations up to $\beta_{max} = 8$, and kept constant for at least the last 15 iterations before i_{max} so the density field can settle at fixed projection sharpness.

At high β, the derivative in (34) scales with β near $\tilde{x}_e = \eta_H$, which can cause period-2 oscillations in the OC update. To damp these oscillations, the filtered sensitivity is averaged between consecutive steps whenever β is constant:

$$\left(\frac{\partial \tilde{F}}{\partial x_e}\right)^{(i)}_{stab} = \gamma \left(\frac{\partial \tilde{F}}{\partial x_e}\right)^{(i)} + (1-\gamma)\left(\frac{\partial \tilde{F}}{\partial x_e}\right)^{(i-1)} \tag{36}$$

with $\gamma = 0.30$ (the history is reset when β increases). At the same time, the OC exponent η_{OC} and move limit ζ in (31) are reduced as β grows ($\eta_{OC} = 0.5$ for $\beta < 2$, $\eta_{OC} = 0.3$ for $2 \le \beta < 8$, $\eta_{OC} = 0.25$ for $\beta = 8$; $\zeta = 0.20, 0.10, 0.05, 0.02$ for $\beta = 1, 2, 4, 8$).

Because a few grey elements along the boundaries can keep oscillating at $\beta = 8$ even after the global response has settled, convergence is checked on the objective history once $\beta = \beta_{max}$ for at least $W = 10$ iterations:

$$\frac{\max_{k\in[i-W,i]} C^{(k)} - \min_{k\in[i-W,i]} C^{(k)}}{C^{(i)}} < \epsilon_C \tag{37}$$

$$\frac{\max_{k\in[i-W,i]} \lambda_1^{(k)} - \min_{k\in[i-W,i]} \lambda_1^{(k)}}{\lambda_1^{(i)}} < \epsilon_\lambda \tag{38}$$

with $\epsilon_C = 10^{-3}$ and $\epsilon_\lambda = 5 \cdot 10^{-3}$. Using the full range ($\max - \min$) over the window avoids false stops when an oscillation has period 2.

7 Buckling stabilization and trade-off formulation

From (18) and (19), choosing $p_K = p_G = 3$ with $E_{min} = 10^{-9}E_0$ means that in low-density elements ($\hat{x}_e \sim 10^{-2}$, where $\hat{x}_e^3 \gg 10^{-9}$) the ratio $E_G(\hat{x}_e)/E_K(\hat{x}_e)$ stays close to 1. The eigenvalue solver can then return spurious buckling modes localized in the void regions [1, 7]. To remove these modes without changing the cubic law in solid members ($\hat{x}_e \to 1$), a rational cutoff $\psi(\hat{x}_e)$ is multiplied into the geometric stiffness interpolation:

$$\psi(\hat{x}_e) = \frac{\hat{x}_e^3}{\hat{x}_e^3 + x_0^3}, \quad E_G(\hat{x}_e) = E_0 \hat{x}_e^3 \psi(\hat{x}_e) \tag{39}$$

with cutoff threshold $x_0 = 0.10$. For $\hat{x}_e \ll x_0$, $E_G(\hat{x}_e) \propto \hat{x}_e^6$, which drives E_G/E_K to zero in the void. The derivative required in (29) is:

$$\frac{\partial E_G(\hat{x}_e)}{\partial \hat{x}_e} = 3E_0 \hat{x}_e^2 \psi(\hat{x}_e)\,(2 - \psi(\hat{x}_e)) \tag{40}$$

In the solver, the first $n_{Eig} = 6$ reciprocal eigenvalues $\mu_m = 1/\lambda_m$ (normalized with $\phi_m^T K \phi_m = 1$) are extracted by passing the sparse LU factorization of $[K]$ as a linear operator to ARPACK. Treating the reference scale μ_1 in the

exponent of (23) as locally constant within each step, the aggregated sensitivity is obtained as a weighted sum of the modal gradients (28):

$$w_m^{KS} = \frac{\exp\left(\rho_{KS}\frac{\mu_m-\mu_1}{\mu_1}\right)}{\sum_{k=1}^{n_{Eig}}\exp\left(\rho_{KS}\frac{\mu_k-\mu_1}{\mu_1}\right)}, \qquad \frac{\partial \mu_{KS}}{\partial \hat{x}_e} = \sum_{m=1}^{n_{Eig}} w_m^{KS}\frac{\partial \mu_m}{\partial \hat{x}_e} \tag{41}$$

with $\rho_{KS} = 20$.

In tension members and near the point load, $\frac{\partial \mu_{KS}}{\partial \hat{x}_e}$ can be positive. When α is large, a positive buckling sensitivity can make the combined gradient in (27) positive, which sets $B_e \leq 0$ in (32) and removes elements from the tension chord. To avoid this, positive buckling sensitivities are clamped to zero before combining them with compliance:

$$\bar{g}_{\mu,e}^{(i)} = \min\left(0, \frac{\partial \mu_{KS}^{(i)}}{\partial \hat{x}_e}\right) \tag{42}$$

The user parameter $\alpha \in [0, 1]$ is mapped to the effective weight $\alpha_{eff}^{(i)}$ ramped over the first $N_{ramp} = 15$ iterations:

$$\alpha_{eff}^{(i)} = 0.65\,\alpha \min\left(1, \frac{i}{N_{ramp}}\right) \tag{43}$$

The factor 0.65 keeps at least 35% weight on compliance even at $\alpha = 1$, while the initial ramp lets the primary load path form before the buckling term reaches full weight.

8 Stress recovery, numerical results, and limitations

At the end of the optimization, the plane-stress vector is computed at the centroid of each element from the nodal displacements u_e:

$$\sigma_e = \begin{bmatrix}\sigma_x \\ \sigma_y \\ \tau_{xy}\end{bmatrix} = E_0[C_0][B_0]u_e \tag{44}$$

From σ_e, the in-plane principal stresses $\sigma_{I,II}$ and the signed dominant principal stress σ_{signed} are obtained as:

$$\sigma_{I,II} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}, \quad \sigma_{signed} = \begin{cases}\sigma_I & \text{if } |\sigma_I| > |\sigma_{II}| \\ \sigma_{II} & \text{otherwise}\end{cases} \tag{45}$$

In the post-processing view, void elements ($\hat{x}_e < 0.40$) are masked out so that tension chords ($\sigma_{signed} > 0$) and compression struts ($\sigma_{signed} < 0$) can be inspected across the sweep.

The solver was tested on a 400×200 mm cantilever domain discretized with 200×100 elements ($r_{min} = 2.0$ elements, $V_f = 0.40$, $E_0 = 210$ GPa, $\nu = 0.3$, thickness $t = 1$ mm, downward tip load $F = 1000$ N). Sweeping α from 0 to 1 in steps of 0.1 increases the fundamental buckling load factor λ_1 by more than a factor of three compared to the pure compliance design ($\alpha = 0$), while compliance C rises accordingly. As α grows, material moves from the tension chord toward the compressed lower flange and internal bracing bars, which divides long compressed struts into shorter segments. For the pure compliance case ($\alpha = 0$), the truss layout matches the reference cantilever topology produced by commercial software (Altair Inspire) under the same boundary conditions and volume fraction; for $\alpha > 0$, commercial packages that only enforce global displacement or stress constraints do not reproduce the secondary buckling bracing observed in the sweep.

The present model is restricted to two-dimensional plane stress (so out-of-plane buckling is not captured) and relies on the linearized pre-buckling assumption, which neglects finite rotations prior to instability. In addition, the dynamically scaled OC update is heuristic: replacing OC with the Method of Moving Asymptotes (MMA) would allow buckling and volume to be treated as separate constraints without manual sensitivity clamping.

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